

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (Biotechnology) (Credit System) Degree Examinations
May - June 2023

19BTC214 - SCIENCE AND ENGINEERING OF BIOMATERIALS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Draw the stress-strain relationship for a ductile biomaterial and explain the various regions / components of the stress-strain graph.	10	L*1	1	1,5,7
	b) Categorise the main class of materials and explain the structure, organisation and properties of titanium and its alloys.	10	L2	1	1,5,7
2.	a) Classify the various sterilisation methods and explain in detail the use of radiation for sterilising medical devices.	10	L2	1	1,5,7
	b) Identify polymeric biomaterial's degradation routes and describe hydrolytic-mediated degradation with suitable examples.	10	L3	1	1,5,7
Unit – II					
3.	a) How would you structurally characterize a biomaterial surface? Explain.	10	L2	2	1,5,7
	b) A biomaterial surface was modified using Plasma irradiation to change its surface wettability. Choose an appropriate characterisation technique to determine the change in wettability and explain the instrument's working with suitable illustrations.	10	L3	2	1,5,7
4.	a) Describe the role of protein adsorption on biomaterials surfaces and its effects on cell adhesion	10	L2	3	3,5,6,7
	b) Explain in detail about Blood - Material Interactions.	10	L2	2	1,5,7
5.	a) How would you fabricate the patient specific scaffolds?	10	L3	2	1,5,7
	b) Explain in detail about transdermal drug delivery.	10	L2	3	3,5,6,7
Unit – III					
6.	a) Discuss the mechanism of bioactivity of bioglass ceramic with suitable diagrams.	10	L2	4	5,7
	b) Classify the different types of dental Implant designs and discuss the different types of metals used in the fabrication of dental implants.	10	L2	4	5,7
7.	a) Explain the composition and structure of cartilage and write a note on synthetic biomaterials for cartilage tissue engineering.	10	L2	4	5,7
	b) Elaborate on various skin substitutes used in plastic surgery.	10	L2	5	5,6,7
8.	a) What are nanoparticles and their unique properties? Categories nanoparticles and elaborate on magnetic nanoparticles.	10	L2	5	5,6,7
	b) Describe the rationale, vectors and process for producing genetically engineered synthetic silk as a biomaterial.	10	L2	5	5,6,7

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
